



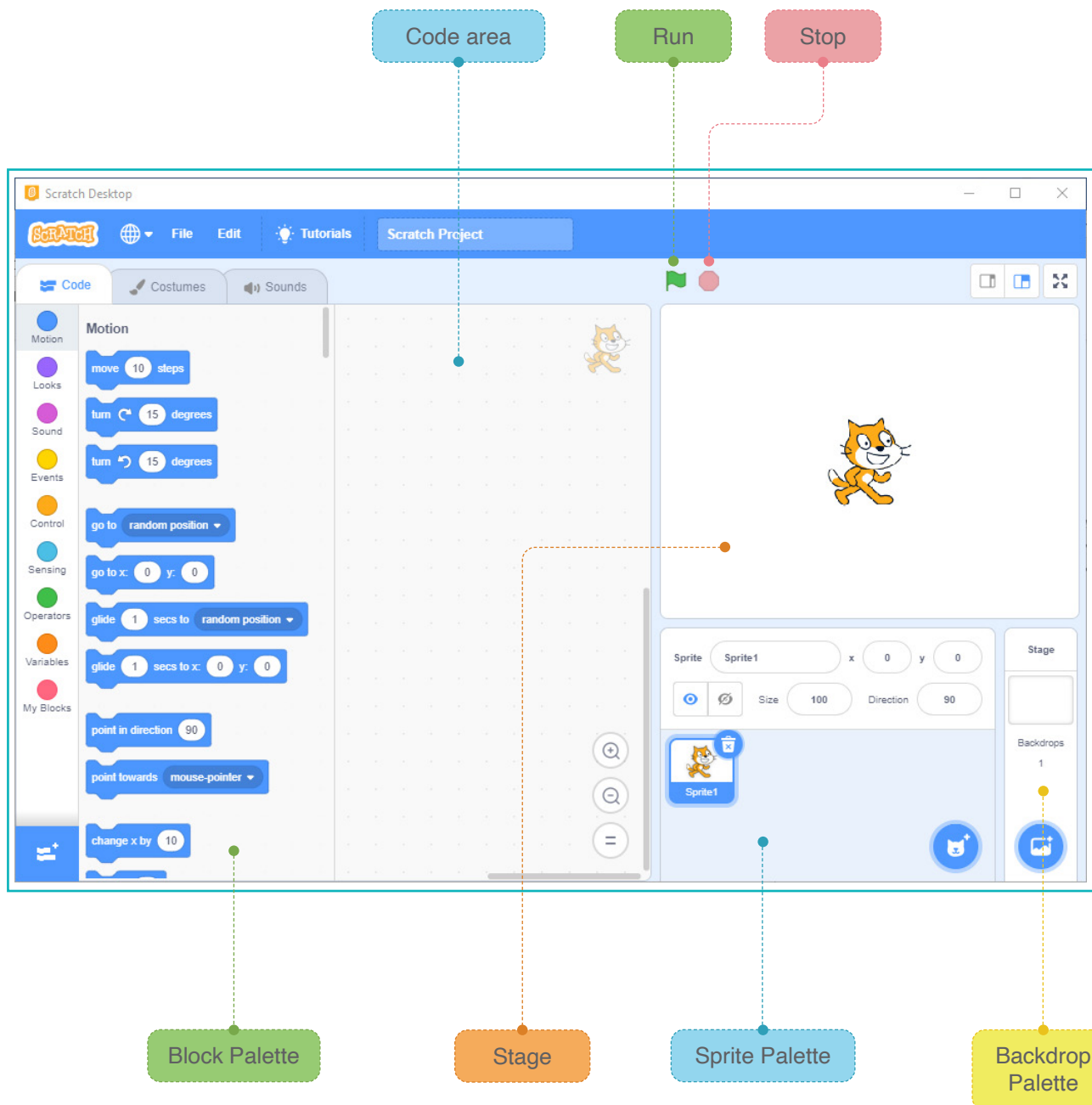
Hello Scratch

03



Scratch: It is a software that uses **blocks** in a **particular order** to give instructions. The instructions can be used to create an animation, story, or game. **Sprites** and **Codes** are used as operating tools in Scratch.

3. Hello Scratch



User Interface of Scratch 3

Sprite

A sprite can be an animated character, a player in a game, or any other object. It is controlled with codes. In Scratch, the **cat** sprite is the '**default sprite**'. We can replace the cat sprite with any different sprite.

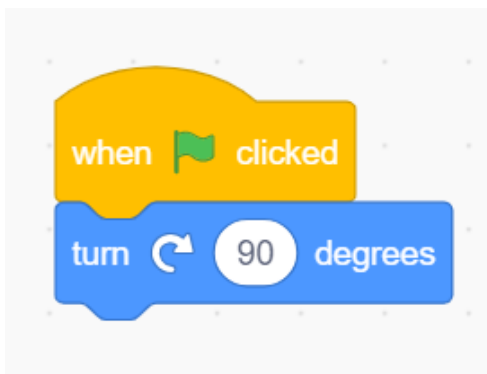


Default sprite

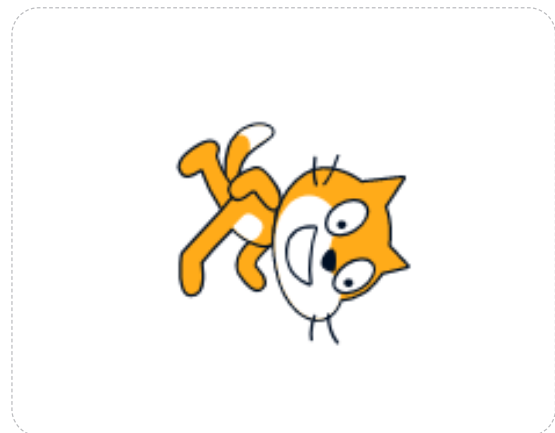
Code

A code is a set of instructions given to a computer to perform an action. Scratch uses a block-based code. Blocks are instructions which are arranged in a sequence to perform a task successfully. We can add codes for a sprite to perform a task. One animation can have multiple sprites. Each sprite can have its own code to perform various actions.

For example: The instruction here is to turn the cat sprite. This kind of instruction is called a code in Scratch.



Code



Result

Bytes

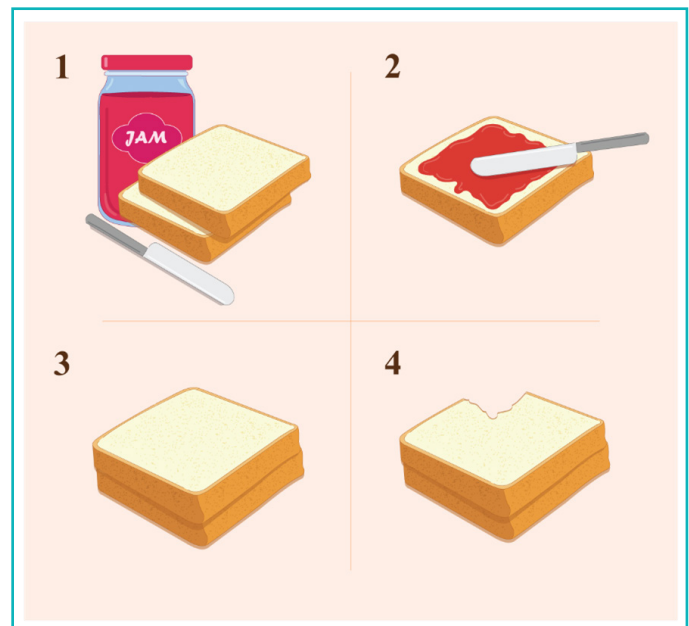
“Default” refers to the set of instructions used by a computer. This will always be used unless the user provides other instructions.

Sequence

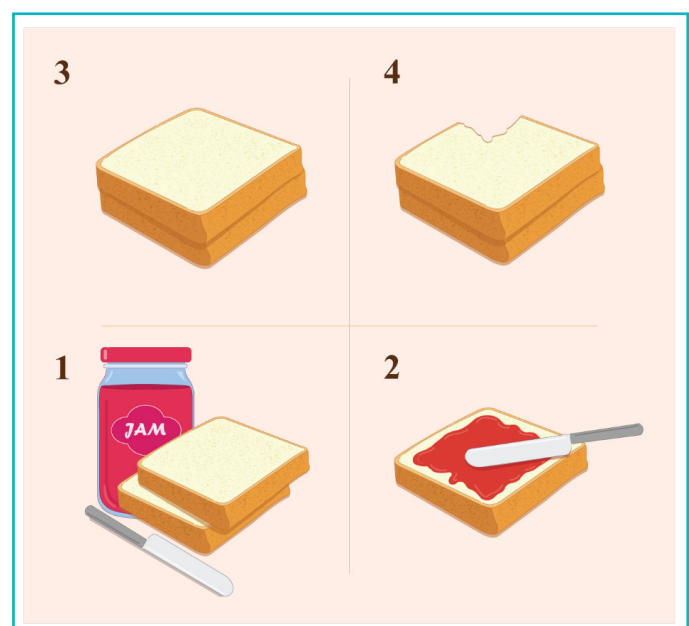
A sequence is an arrangement of two or more things in a particular order. A task is successfully completed if we follow the correct sequence required to perform it.

For example: To make a jam sandwich, there is a sequence you will have to follow. This sequence will be:

1. Take 2 slices of bread and some jam.
2. Spread the jam on one slice.
3. Place the second slice on top of the first slice.
4. Eat your sandwich.

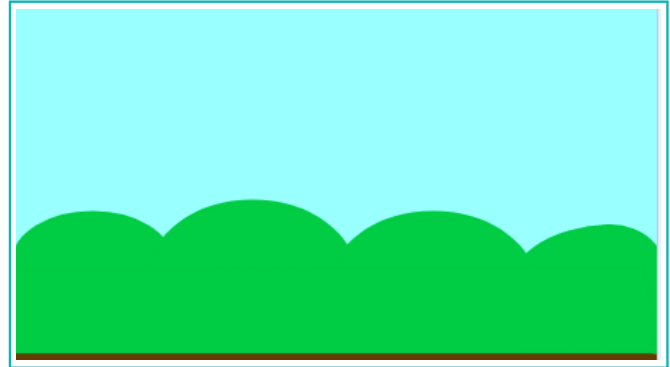


But, if we change the sequence, it will not work.



Backdrop

A backdrop is an image that can be used as a background. A backdrop is added to the stage to give an appropriate background to an animation, game, or story.



Example of backdrop

Stage

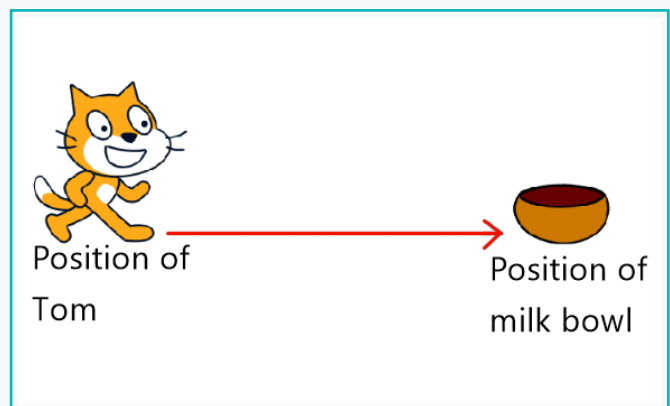
The Stage is the display screen where we can watch the sprites move.

Activity

Tom the cat is hungry. He wants to drink a bowl of milk. But it is not within his reach. We must move Tom from his current position to the position of the milk bowl. To achieve this output, we will need to use a sequence of codes.

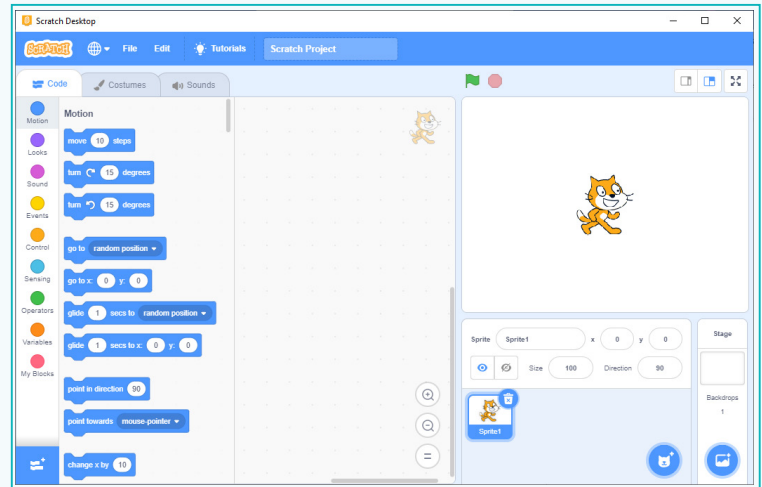
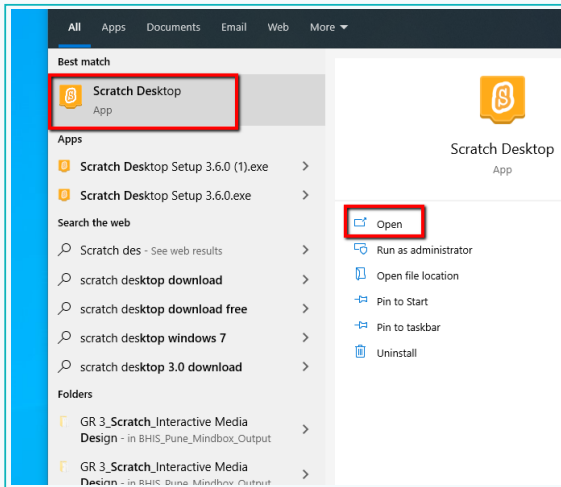
For Tom to reach the bowl of milk, he has to walk through the farm. We will follow the given sequence:

1. Add backdrop: Farm
2. Add Sprite: Default sprite (Tom)
3. Give instructions: Create a code
4. Test the code: Run

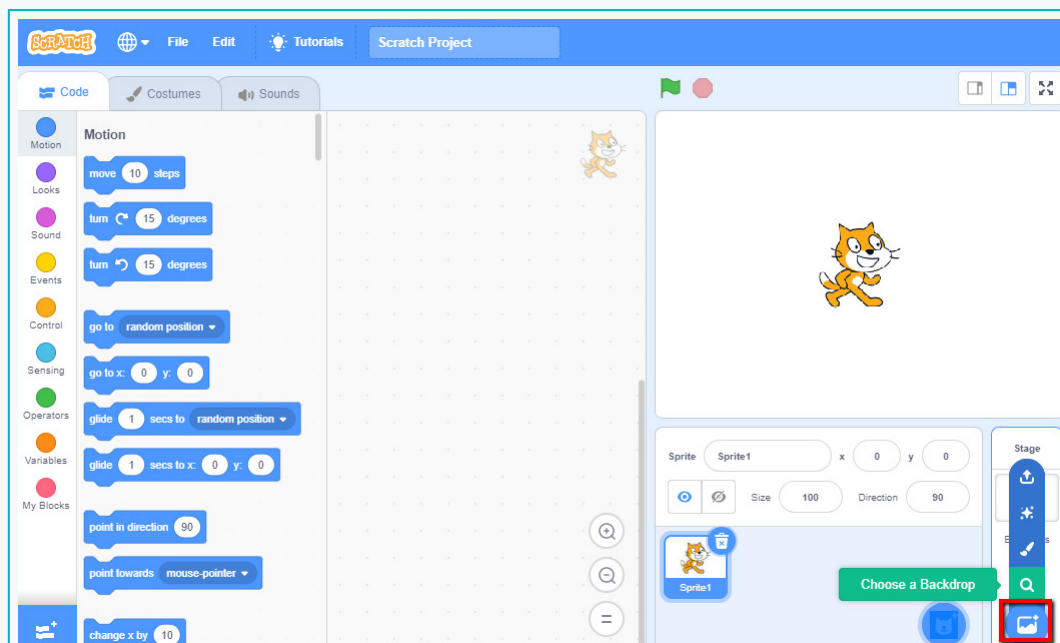


Activity

- 1 To open the software, type **Scratch Desktop** in the search bar.

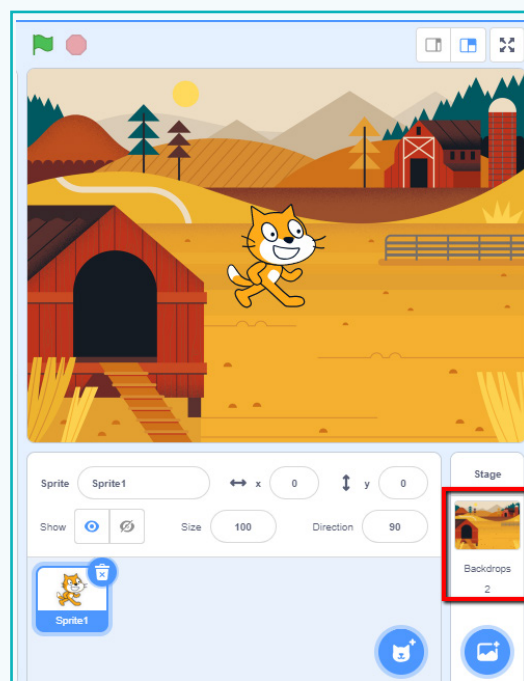
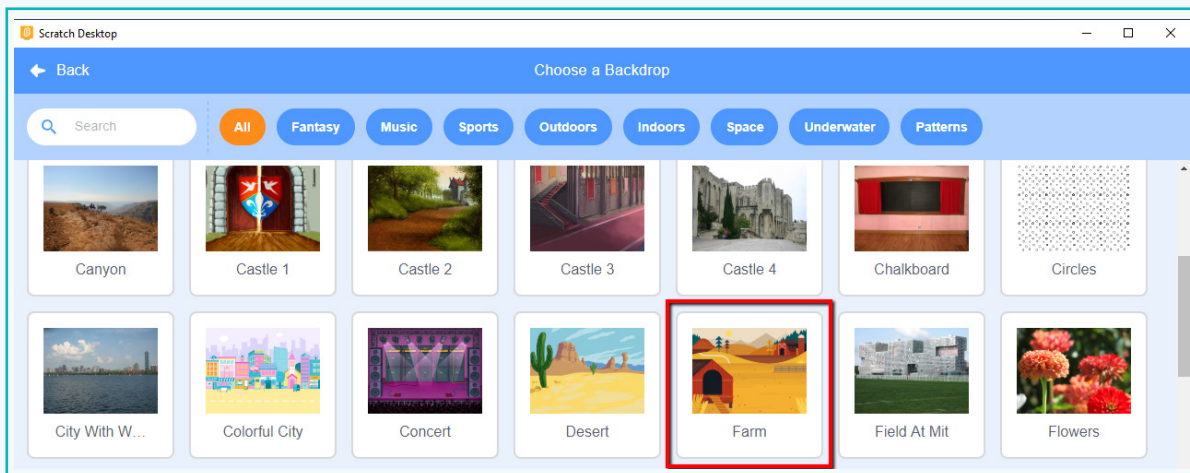


- 2 Click on **Backdrop Palette** and then select **Choose a Backdrop**.



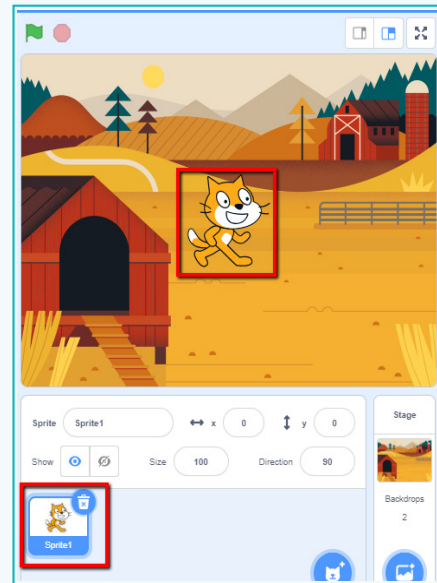
Activity

- 3 A Backdrop library will open in Scratch.
- 4 Choose the Farm backdrop.

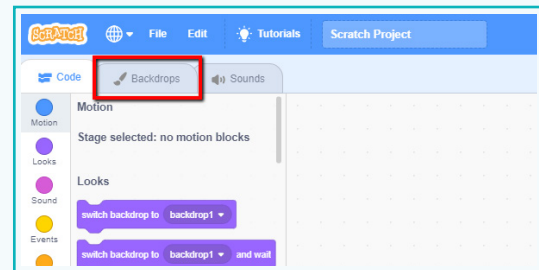


Activity

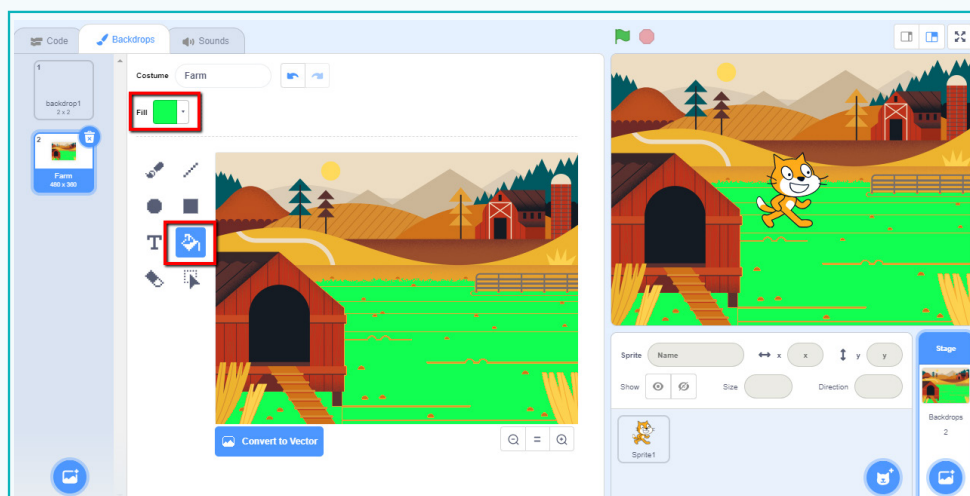
We will use the default sprite of the cat. Let's call him 'Tom'. Tom matches the colour of the backdrop, so it is hard to see him. We will change the colour of the backdrop to Green so we can see Tom clearly.



- 5 To change the colour of the farm, select the backdrop and click on **backdrops**.

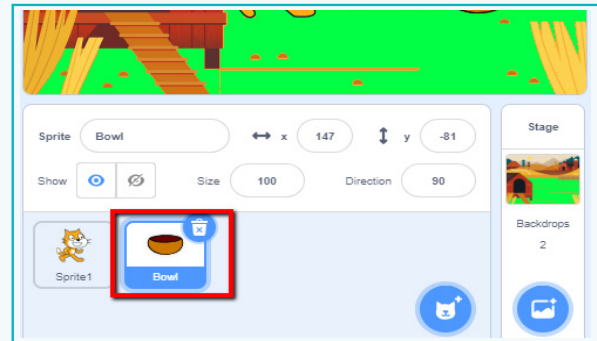


- 6 To fill green colour in the farm backdrop, select Colour and use the **Fill Tool**.



Activity

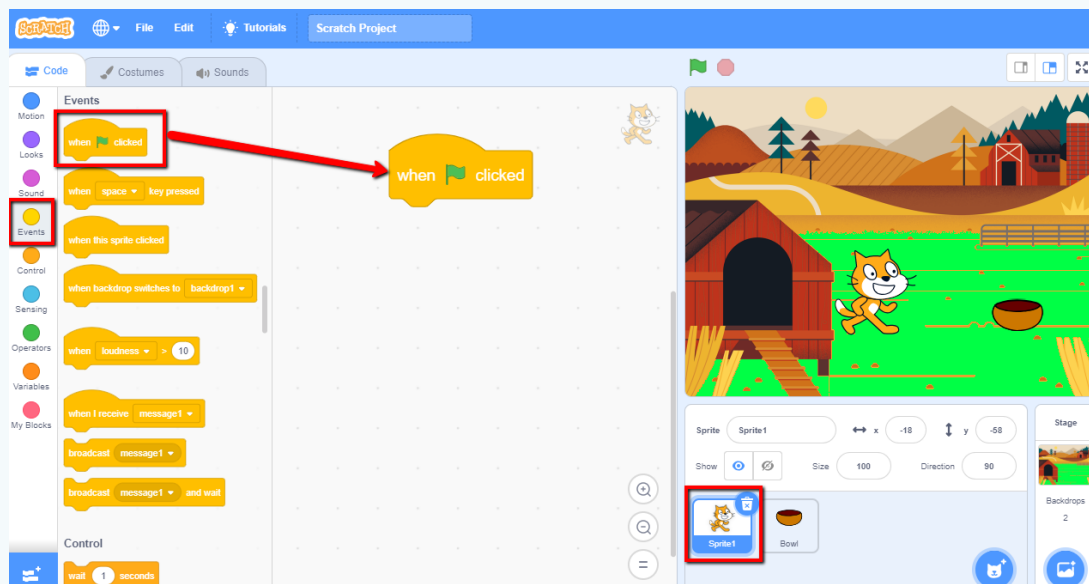
- 7 Click on **code** and go back to the main screen.
- 8 Select **bowl** from the sprite library. Line up both the sprites opposite to each other.



Looks like Tom needs move at least 50 steps to reach the milk bowl. To help Tom reach the bowl, we add a movement block with a value of 50 steps.

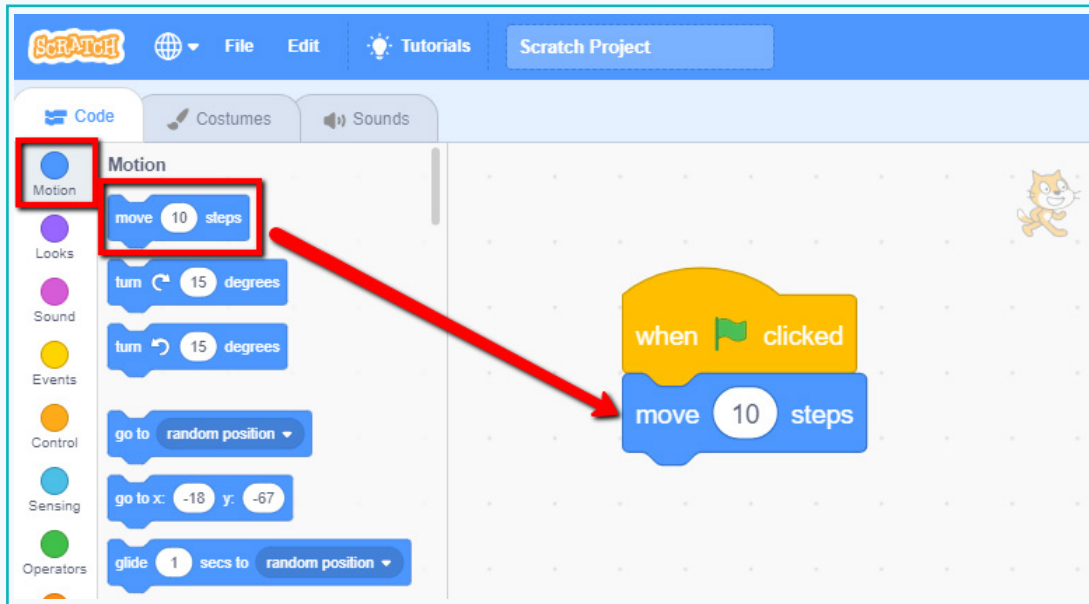
- 9 Add **movement blocks** to move the cat sprite towards the bowl of milk. Let's give a value of 50 **steps** to the cat sprite.
- 10 From the Events block, drag and drop a “**When flag clicked**”  block to the code area.

The **Green Flag** is a programming feature. So that, “when flag clicked”, it will run all the scripts that are placed under it.

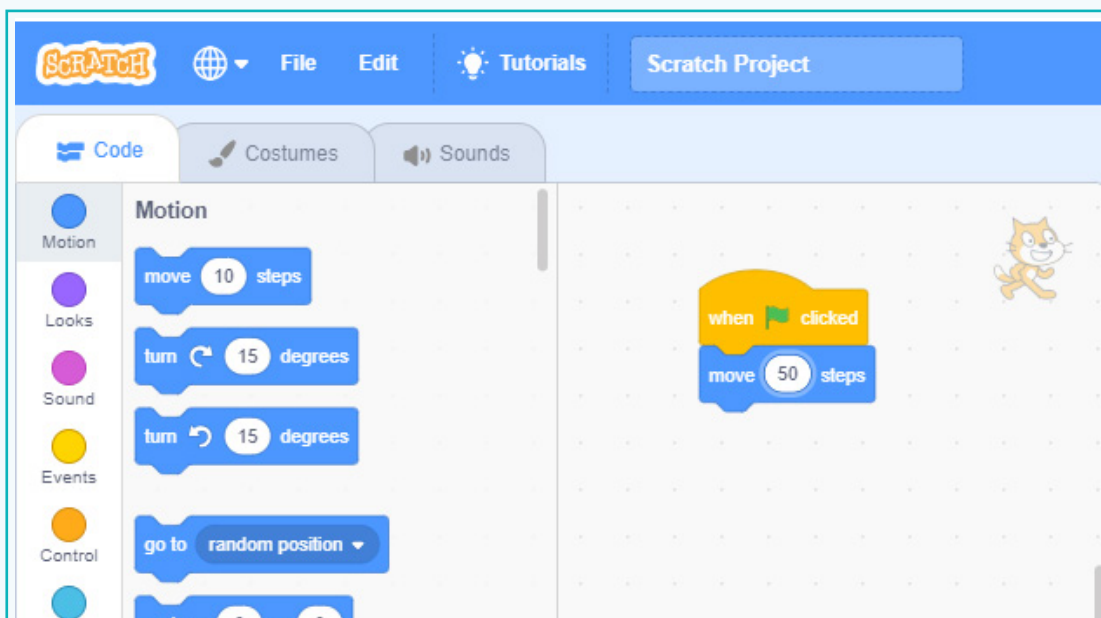


Activity

- 11 Drag and drop the “**Move 10 steps**” block from the **Motion** block to the code area.

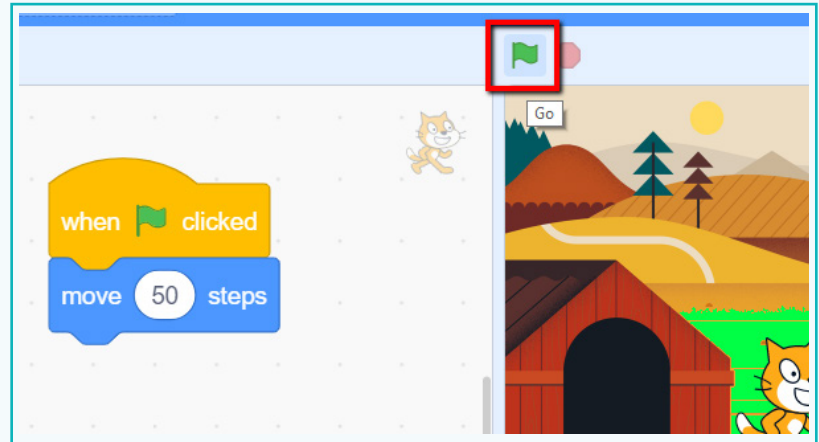


- 12 Double click on the number and change it to **50**.



Activity

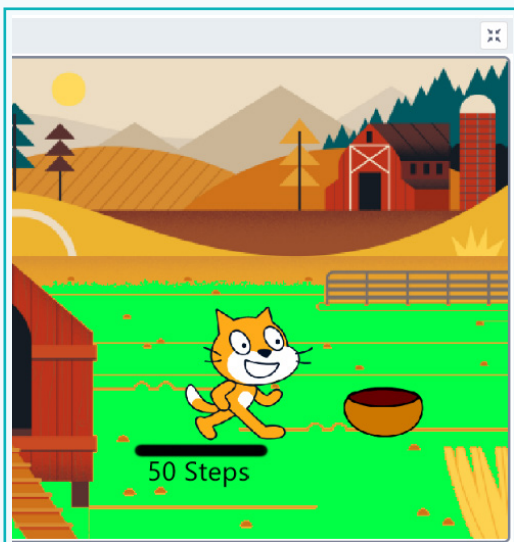
- 13 To see the movement, click on the Green Flag .



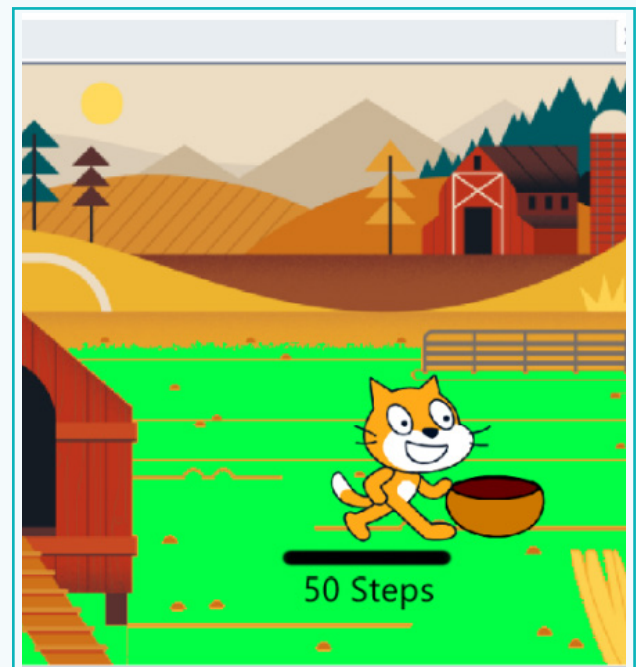
- 14 Observe the movement of Tom, the cat sprite, on the stage.

Tom is unable to reach the bowl of milk in just 50 steps.

- 15 Make him move another **50 steps** to reach the bowl.

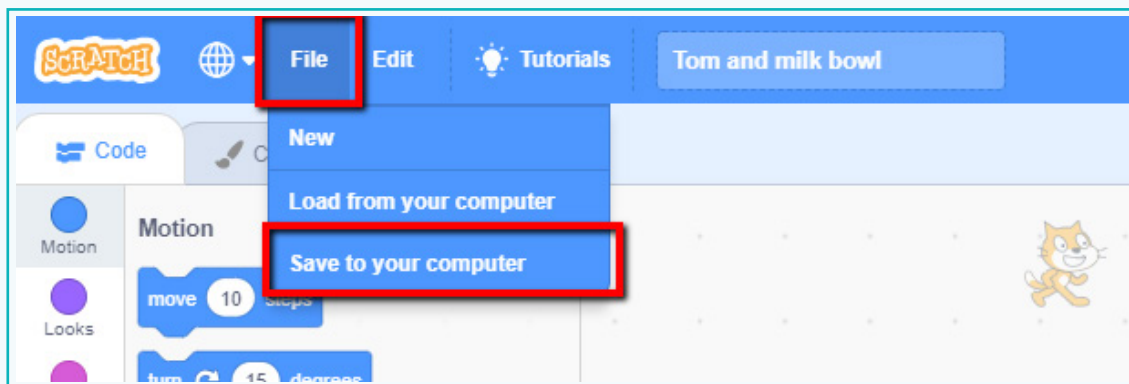


Tom has now reached the bowl of milk!

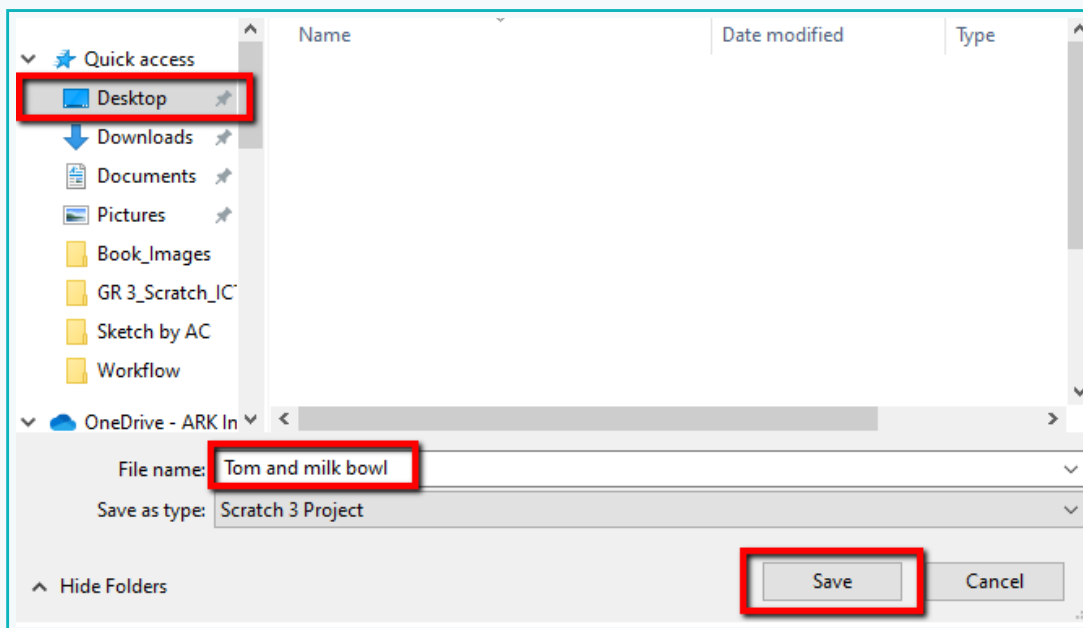


Activity

- To save the file:
- 16 Click on **File** and select **save to your computer**.



- 17 Name the file “**Tom and milk bowl**”.
- 18 Select Desktop and click on Save.

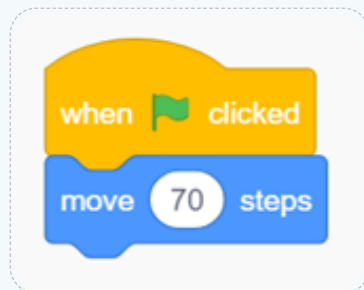
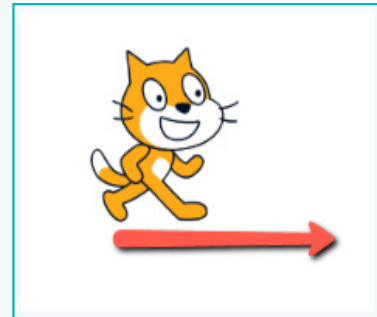


We have used Motion and Events blocks to move Tom to the bowl of milk through a farm.

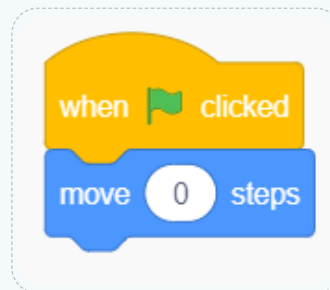
Exercise

Choose the Correct Answer

- 1 Which option will move the sprite in a forward direction?

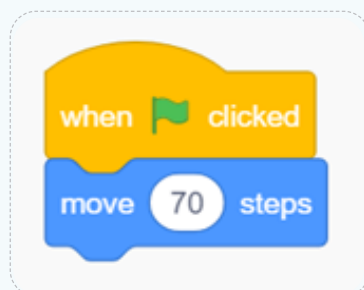
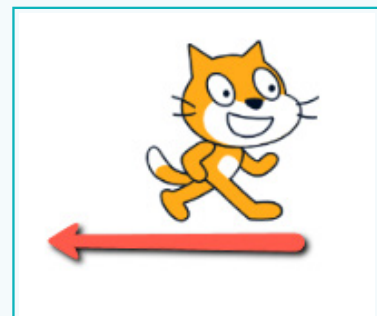


a. _____

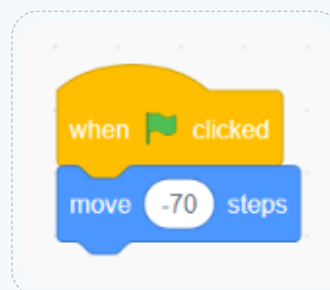


b. _____

- 2 Which option will move the sprite in a backward direction?



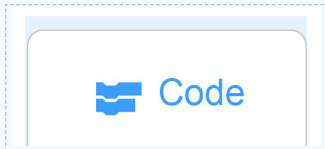
a. _____



b. _____

Exercise

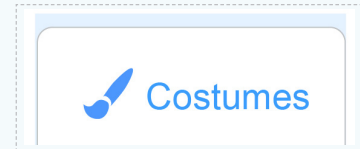
- 3 Which option allows us to change the colour of the backdrop?



a. _____



b. _____



c. _____

State whether true or false

- 4 A Sprite can be an animated character.

▲ _____

- 5 A Code is used to control the action of a sprite.

▲ _____

- 6 A Stage is used to give instructions.

▲ _____



Tech Flash

- Scratch** : Scratch is a software that uses blocks in a sequence to give instructions to create an animation, story, or game.
- Sprite** : A sprite can be an animated character, a player in a game, or any other object.
- Code** : A code is a set of commands given to perform an action in a program. Scratch uses a block-based code.
- Stage** : A stage is used to display the animation, game, or story.
- Flood Fill** : Flood fill is used to fill colour in a drawing.